

Bărbuceanu Alin-Andrei

Student

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🇷🇴 Romanian 🌐 alin-andrei barbuceanu

PROFILE

Second-year ETTI student, passionate about programming, technology, and continuous growth. I am organized, adaptable, and always eager to learn new things and develop my skills. I communicate clearly and work well in collaborative environments, keeping a positive and solution-oriented mindset. Once I start a task, I stay focused and committed to delivering the best possible result, aiming for both quality and constant improvement.

EDUCATION

Bachelor's Degree – Electronics, Telecommunications and Information Technology (ETTI)

09/2024 – 07/2028
Bucharest, Romania

University Politehnica of Bucharest 🎓

Coursework focused on programming fundamentals, electronics, circuits, and computer systems. Developing skills in C/C++, Python, and problem solving through hands-on labs and technical assignments. Currently building my software development skills through a Web Development course and expanding my networking knowledge with CCNA, while connecting everything with my electronics background.

PROJECTS

QR Code Generator & Scanner

Python · Tkinter · OpenCV

Developed a desktop application that generates and scans QR codes. Used Tkinter for the interface, OpenCV for scanning from images or webcam, and the “qrcode” library for generation. Organized the project into modules (UI, backend, image processing) to make it easy to extend. Learned how to integrate a GUI with real-time camera input and image processing.

Blackjack Game – ESP32

C Programming · ESP32 · OLED Display · Hardware Integration

Developed a fully functional Blackjack game on an ESP32 board, using a 128×32 OLED display, physical buttons, LEDs, and a buzzer for feedback. Implemented card drawing logic, score calculation, game states (hit, stand, bust), and real-time interaction with the user. The project helped me understand embedded programming, handling I/O peripherals, timing, and integrating hardware components with game logic.

TodoMVC Web Application

HTML · CSS · JavaScript

Built a fully functional To-Do web app where users can add, edit, mark, and delete tasks. Implemented automatic data persistence using LocalStorage. Strengthened my understanding of DOM manipulation, event handling, and structuring a small but complete JavaScript application. Focused on clear logic and a clean, simple UI.

Inductive Metal Detector

Analog Electronics · LC Oscillator · BC547

Built an inductive metal detector using an LC oscillator to observe frequency changes when metal approaches the coil. The circuit includes two BC547 transistors, a hand-wound ~40-turn coil, a 10 kΩ potentiometer, a 1 kΩ resistor, several capacitors (10 nF, 22 nF, 100 nF), and an active buzzer, all powered at 5V. This project helped me understand practical oscillator operation, frequency variation, signal amplification, and breadboard debugging.

SKILLS

Programming

- C / C++ — Intermediate
- Python — Intermediate
- HTML / CSS — Intermediate
- JavaScript — Beginner

Engineering & Technical Tools

- Electronic Circuit Design — Intermediate
- Breadboard Debugging — Intermediate
- TINA-TI — Beginner
- Cadence Design Tools — Beginner
- MATLAB — Beginner
- Cisco Packet Tracer — Beginner

Software

- Microsoft Office — Advanced

CERTIFICATES

JavaScript Fundamentals Certificate

Issued by: HackAcademy

Cambridge English Certificate

Issued by: Cambridge Assessment English

IC3

Issued by: Certiport / Pearson VUE

LANGUAGES

- Romanian — Native
- English — C1

COURSES

Web Development Course — HTML, CSS, JavaScript

Completed at: HackAcademy

CCNA Fundamentals — Subnetting, routing, addressing, Packet Tracer

In progress at: InfoAcademy